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MAST20030: Differential Equations
Tutorial Solutions: Week 8

Solution to Question 1.

(a) $L = 2$, i.e., the period is 4.

(b) In terms of piecewise functions:

$$f(t) = \begin{cases} 0 & -2 < t \leqslant -1 \\ t+1 & -1 < t \leqslant 0 \\ 1-t & 0 < t \leqslant 1 \\ 0 & 1 < t \leqslant 2 \end{cases}$$

and in terms of Heaviside functions

$$f(t) = (t+1)H_{-1}(t) - (t+1)H_0(t) + (1-t)H_0(t) - (1-t)H_1(t) = (t+1)H_{-1}(t) - 2tH_0(t) - (1-t)H_1(t).$$

(c) The function is even so the sin terms will be zero.

$$\begin{aligned} a_0 &= \frac{1}{2}\left(\int_{-1}^{0} t+1\,\mathrm{d}t + \int_0^1 1-t\,\mathrm{d}t\right) = \frac{1}{2} \\ a_n &= \frac{1}{2}\left(\int_{-1}^{0}(t+1)\cos\frac{n\pi t}{2}\,\mathrm{d}t + \int_0^1 (1-t)\cos\frac{n\pi t}{2}\,\mathrm{d}t\right) = \int_0^1 (1-t)\cos\frac{n\pi t}{2}\mathrm{d}t \\ &= \left[\frac{2}{n\pi}\sin\frac{n\pi t}{2}\right]_0^1 - \left[\frac{2t}{n\pi}\sin\frac{n\pi t}{2}\right]_0^1 + \int_0^1 \frac{2}{n\pi}\sin\frac{n\pi t}{2}\mathrm{d}t \\ &= \left[-\frac{4}{n^2\pi^2}\cos\frac{n\pi t}{2}\right]_0^1 = \frac{4}{n^2\pi^2}\left(1-\cos\frac{n\pi}{2}\right). \end{aligned}$$

And so the Fourier series is given by

$$F(t) = \frac{1}{4} + \sum_{n=1}^{\infty}\frac{4}{n^2\pi^2}\left(1-\cos\frac{n\pi}{2}\right)\cos\frac{n\pi t}{2}.$$

Optional to simplify outside of class: If n is odd, i.e., $n = 2m-1$ then $\cos\dfrac{(2m-1)\pi}{2} = 0$ and $a_n = \dfrac{1}{n^2\pi^2} = \dfrac{4}{(2m-1)^2\pi^2}$. If n is even, i.e., $n = 2m$ and m is even, i.e., $n = 4k$, then $\cos\dfrac{4k\pi t}{2} = 1$ and $a_n = 0$. If n is even and m is odd, i.e., $n = 4k-2$, then $\cos\dfrac{(4k-2)\pi}{2} = -1$ and $a_n = \dfrac{8}{(4k-2)^2\pi^2} = \dfrac{2}{(2k-1)^2\pi^2}$ giving

$$F(t) = \frac{1}{4} + \sum_{n=1}^{\infty}\frac{4}{(2n-1)^2\pi^2}\cos\frac{(2n-1)\pi t}{2} + \sum_{n=1}^{\infty}\frac{2}{(2n-1)^2\pi^2}\cos(2n-1)\pi t\,.$$

(d) For large n the non-zero coefficients scale as $\dfrac{1}{n^2}$ so we expect reasonably fast convergence.

(e) As $\left|1-\cos\dfrac{n\pi}{2}\right| \leqslant 2$ we have that

$$\left|\frac{4}{n^2\pi^2}\left(1-\cos\frac{n\pi}{2}\right)\cos\frac{n\pi t}{2}\right| \leqslant \frac{8}{n^2\pi^2}.$$

Now since $\displaystyle\sum_{n=1}^{\infty}\frac{8}{n^2\pi^2}$ converges, the Fourier series, $F(t)$, converges uniformly.

Solution to Question 2.

(a) Assuming that we can integrate this Fourier series term-by-term, we have

$$x^2 = \int_0^x 2t\,\mathrm{d}t = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}\int_0^x \sin(\pi nt)\,\mathrm{d}t = \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(\pi nt)\Big|_0^x$$
$$= \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(\pi nx) - \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}, \qquad -1 < x < 1.$$

Since this is the Fourier series of x^2, the constant term is $\frac{1}{2}a_0 = \frac{1}{2}\int_{-1}^{1} x^2\,\mathrm{d}x = \frac{1}{3}$. Thus, we have

$$x^2 = \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(\pi nx) + \frac{1}{3}, \qquad -1 < x < 1.$$

Because the periodic extension of x^2, $-1 \leqslant x \leqslant 1$, is continuous, this equality holds for $x = \pm 1$ as well.

(b) This time we calculate as follows:

$$x^3 = \int_0^x 3t^2\,\mathrm{d}t = \frac{12}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\int_0^x \cos(\pi nt)\,\mathrm{d}t + \int_0^x 1\,\mathrm{d}t = \frac{12}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3}\sin(\pi nx) + x, \qquad -1 < x < 1.$$

While we don't have any constant term to evaluate, this is *not* a Fourier series because of the term x. To arrive at the Fourier series of x^3, we simply substitute that of x:

$$x^3 = \frac{12}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3}\sin(\pi nx) - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin(\pi nx), \qquad -1 < x < 1.$$

But, this spoils the desired decay property of the Fourier coefficients (they now decay as $\frac{1}{n}$, not $\frac{1}{n^3}$). To get the desired decay property, we simply subtract the term x (*ie.*, $\alpha = 1$):

$$x^3 - x = \frac{12}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3}\sin(\pi nx), \qquad -1 < x < 1.$$

(c) Integrating the Fourier series of $x^3 - x$ once more will give a Fourier series whose coefficients decay as $\frac{1}{n^4}$ because there is no constant term to spoil things. We therefore obtain

$$x^4 - 2x^2 = \int_0^x 4(t^3 - t)\,\mathrm{d}t = \frac{48}{\pi^4}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^4}\cos(\pi nx) + \frac{48}{\pi^4}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^4}, \qquad -1 < x < 1.$$

This establishes that $\beta = 2$ and we evaluate the constant term as before: $\frac{1}{2}a_0 = \frac{1}{2}\int_{-1}^{1}(x^4 - 2x^2)\,\mathrm{d}x = -\frac{7}{15}$. This gives us two results:

$$\sum_{n=1}^{\infty}\frac{(-1)^n}{n^4} = -\frac{7\pi^4}{720} \qquad \text{and} \qquad x^4 - 2x^2 = -\frac{7}{15} + \frac{48}{\pi^4}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^4}\cos(\pi nx), \quad -1 < x < 1.$$

To evaluate the final sum, note that the periodic extension of $x^4 - 2x^2$ is continuous, hence that the latter result also holds for $x = \pm 1$. We may therefore substitute $x = 1$ to obtain

$$-1 = -\frac{7}{15} - \frac{48}{\pi^4}\sum_{n=1}^{\infty}\frac{1}{n^4} \;\Rightarrow\; \sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}.$$